

Notes: Griffin, Hirschey, and Kelly (RFS, 2011)

How Important Is the Financial Media in Global Markets?

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1 Big picture

- **Question.** How important is the financial media for stock-price discovery around the world, and why does the market reaction to public firm news differ so much across countries?
- **Setting.** The paper studies 870,000-plus firm news articles from Factiva, covering 2,593 firms in 56 markets from January 2003 to June 2009. The sample includes 26 developed markets and 30 emerging markets.
- **Core empirical object.** The paper asks whether stocks move more on public-news days than on non-news days. It measures this first around a common and comparable event – earnings announcements – and then around all firm-level media articles.
- **Main finding.** In most developed markets, stocks move much more on days with public news. In many emerging markets, volatility on public-news days is similar to volatility on non-news days. The contrast is large: developed-market earnings-announcement days have about 38% higher absolute abnormal returns than normal days, while emerging markets have about 15% higher absolute abnormal returns.
- **Puzzle.** Emerging-market firms receive less news coverage, so one might expect each public news item to be especially informative. The opposite is often true: public media news explains less of stock-price movement in emerging markets.
- **Candidate explanations.** The authors consider four broad mechanisms: public information may arrive earlier through other public channels; insiders may trade before official news becomes public; news transmission may be faster or better in some countries; and accounting quality may make earnings signals more or less reliable.
- **Central interpretation.** The strongest explanation is insider trading. In countries where insider trading is viewed as more prevalent, public announcements are less surprising because part of the information appears to be incorporated into prices before the official public release.
- **Second major explanation.** Technological development, interpreted as the speed and breadth of news dissemination, also matters. In countries with better technology and news transmission, public news is more closely connected to price movements.
- **Important nuance.** Accounting standards help explain reactions to earnings announcements, but not general news. This is intuitive: accounting quality matters most when the news itself is accounting information.
- **Extra evidence.** The paper does not stop at cross-country regressions. It also shows that countries with weak public-news reactions have much larger takeover price run-ups before public merger announcements and smaller return reversals after extreme non-news-day returns. Both patterns are consistent with more private informed trading.
- **Takeaway.** The financial media matters, but it matters very differently across markets. In high-reaction countries, prices move strongly when news becomes public. In low-reaction countries, much of the information appears to enter prices through non-public channels before the media announcement.

2 Data and measurement (key objects)

2.1 Return data and stock sample

Daily stock returns and market capitalizations come from CRSP for the United States and Thomson Financial Datastream for non-U.S. markets. The authors focus on common equity and apply detailed screens to remove preferred stock, warrants, unit trusts, cross-listings, GDRs, duplicates, and other non-common equity securities.

The paper requires stocks to be actively traded: a stock must have nonzero price changes on at least 50% of trading days in the prior year.

Interpretation. This filter is important because the paper studies whether information release moves prices. Very illiquid stocks may show stale prices, mechanically weakening measured news reactions even when information is economically relevant.

2.2 News articles from Factiva

The media data come from the Dow Jones Factiva archive. Factiva contains articles from thousands of sources across many countries and languages. The paper identifies firm news by searching for the firm's name in the headline or lead paragraph of an article.

The final general-news sample contains:

- 2,593 firms,
- 1,342 firms from 26 developed markets,
- 1,251 firms from 30 emerging markets,
- 572,987 articles for developed-market firms,
- 298,614 articles for emerging-market firms,
- 871,601 total news articles.

A firm-year is included only if the firm has at least one news article during the year and no more than 75% of trading days with news, so that there are enough non-news days for comparison.

Economic message. Developed-market firms receive more news: 427 articles per firm on average versus 239 in emerging markets. Yet the key result is not simply that developed markets have more media. It is that their stock prices move more *when* media news arrives.

2.3 Developed versus emerging market classification

The paper classifies a country as developed if its 2005 gross national income per capita is above the World Bank high-income threshold of 10,725 U.S. dollars. Other countries are classified as emerging.

Use in the paper. This distinction is descriptive rather than the final explanation. The main regressions ask which country characteristics explain why some markets react strongly to news and others do not.

2.4 Earnings-announcement sample

Earnings announcements are used because they are common, comparable, and likely to contain value-relevant information. The authors start from Bloomberg earnings-announcement dates and then use Factiva to check whether the date corresponds to the first public release of earnings news.

The automated checking procedure keeps events for which:

- the firm's name appears in the headline or lead paragraph,
- Factiva tags the article as earnings news,
- the headline contains an earnings-related string,
- no earlier article meeting the same criteria appears in the prior 60 days.

The final earnings sample contains 5,583 announcements: 3,504 from developed markets and 2,079 from emerging markets.

Interpretation. This is a major design choice. If event dates are wrong, the measured event-day reaction will be biased downward. The authors therefore trade sample size for date accuracy.

2.5 Merger-announcement sample

For takeover leakage tests, the paper constructs initial merger announcements using the earliest date from SDC, FactSet, and Bloomberg. It also searches Factiva to ensure that no merger-related article appears from 60 calendar days to two trading days before the announcement.

The final strict merger sample has 571 events: 466 from developed markets and 105 from emerging markets.

Interpretation. Takeovers are useful because public anticipation is harder to justify with ordinary public information. A pre-announcement run-up therefore provides a more direct diagnostic for private information leakage.

2.6 Country-level explanatory variables

The paper studies country variables linked to four hypotheses. Important variables include:

- **Insider trading:** a Global Competitiveness Report survey measure, reversed so higher values mean insider trading is more pervasive.
- **Technological development:** a composite of measures of technology availability, readiness, and sophistication.
- **Free press:** survey evidence on whether media can publish without censorship or retaliation.
- **News clustering:** the relative concentration of articles around event windows.
- **In-depth articles:** articles with more than 500 words per firm mentioned.
- **Financial press:** coverage by major business outlets such as Reuters, Dow Jones, *The Wall Street Journal*, and *Financial Times*.

- **Accounting standards:** survey-based strength of accounting standards.
- **International GAAP/IFRS:** the fraction of firms using international accounting standards or U.S. GAAP.

2.7 What is being measured conceptually?

The main measurement idea is simple:

If public news is genuinely informative to investors, then idiosyncratic stock volatility should be higher when that news is released than when it is not released.

The paper therefore does not need to classify the news as good or bad. It studies the absolute price movement around news.

3 Models and empirical specifications (all equations + interpretation)

3.1 Abnormal return and event volatility

Let $AR_{i,t}$ denote the abnormal return for stock/event i on day t . The baseline abnormal return is the stock return minus the local value-weighted market return. The main volatility measure uses absolute abnormal returns, $|AR_{i,t}|$.

For an earnings announcement, the event window is $[-1, 2]$ around the announcement date. Event volatility for country j is

$$Vol_j^{Event} = \frac{1}{N_j} \sum_{i=1}^{N_j} \frac{1}{4} \sum_{t=-1}^2 |AR_{i,t}|. \quad (1)$$

Normal volatility is measured over the surrounding non-event window, $[-55, -2]$ and $[3, 55]$:

$$Vol_j^{Normal} = \frac{1}{N_j} \sum_{i=1}^{N_j} \frac{1}{107} \left(\sum_{t=-55}^{-2} |AR_{i,t}| + \sum_{t=3}^{55} |AR_{i,t}| \right). \quad (2)$$

Interpretation. The event window captures the price movement when the earnings news becomes public. The normal window captures the stock's usual idiosyncratic volatility around nearby non-event days.

3.2 Normalized volatility

The main earnings-reaction measure is

$$Normalized\ Volatility_j = \frac{Vol_j^{Event}}{Vol_j^{Normal}} - 1. \quad (3)$$

A value of 0.50 means that average event-window absolute abnormal returns are 50% larger than normal absolute abnormal returns.

Economic interpretation. This measure makes reactions comparable across countries with different baseline volatility. A country with high ordinary volatility is not automatically classified as a high news-reaction country unless volatility rises around the event.

3.3 Differenced volatility

As a robustness measure, the paper also uses

$$Differenced\ Volatility_j = Vol_j^{Event} - Vol_j^{Normal}. \quad (4)$$

Interpretation. The differenced measure asks how many percentage points of extra absolute return occur during the event window. It is less sensitive to small denominators than a ratio.

3.4 Residual earnings reactions

To check whether country rankings are driven by firm composition, the authors estimate a firm/event-level residual reaction regression of the form

$$Vol_{i,t}^{Event} - Vol_{i,t}^{Normal} = \alpha + Industry_i + \beta_1 \log(MV_{i,t-1}^{USD}) \quad (5)$$

$$+ \beta_2 PriorYearPctPriceTrade_{i,t} + \beta_3 Accuracy_{country} + \beta_4 PriorNewsCoverage_{i,t} + u_{i,t} \quad (6)$$

The residuals are aggregated by country.

Interpretation. This asks whether developed/emerging differences remain after controlling for industry, firm size, trading activity, date-checking accuracy, and prior media coverage.

3.5 Firm-level news regression

For general news, the paper estimates firm-level regressions:

$$\ln(1 + |AR_{i,t}|) = \alpha_i + \beta_{NewsDay,i} NewsDay_{i,t-1} + \beta_{ArtCount,i} ArtCount_{i,t-1} + \varepsilon_{i,t}. \quad (7)$$

Here:

- $NewsDay_{i,t-1}$ indicates whether at least one article about firm i appeared after the prior market close and before the current market open.
- $ArtCount_{i,t-1}$ is the number of such articles.
- The adjusted R^2 measures how much of the stock's absolute abnormal return variation is associated with news days and article counts.

Interpretation. A higher adjusted R^2 means that public media coverage is more tightly connected to stock-price volatility.

3.6 News-day versus non-news-day ranking measures

The paper also compares news-day and non-news-day volatility more directly. For each country, average firm-year volatility observations are ranked. The paper computes the percentile rank of news-day volatility and centers it by subtracting 0.50:

$$RankMeasure_j = AvgPercentileRank(NewsDayVolatility_j) - 0.50. \quad (8)$$

It also computes mean and median ratios:

$$MeanRatio_j = \frac{Mean(NewsDayVolatility_j)}{Mean(NonNewsDayVolatility_j)} - 1, \quad (9)$$

$$MedianRatio_j = \frac{Median(NewsDayVolatility_j)}{Median(NonNewsDayVolatility_j)} - 1. \quad (10)$$

Interpretation. The rank measure is less sensitive to outliers. The mean ratio captures large events. The median ratio captures the typical news day.

3.7 Cross-country regression framework

The cross-country regressions take the broad form

$$Reaction_j = \alpha + \gamma X_j + \eta_j, \quad (11)$$

where $Reaction_j$ is a country-level news-reaction measure and X_j is a country characteristic such as insider-trading prevalence, technological development, free press, or accounting standards.

Many specifications also include GDP per capita:

$$Reaction_j = \alpha + \gamma X_j + \delta GDPpc_j + \eta_j. \quad (12)$$

Interpretation. GDP per capita is a broad development control. A variable that survives GDP per capita is less likely to be merely proxying for rich versus poor countries.

3.8 Composite general-news reaction measure

For general news, the paper combines four standardized measures:

1. adjusted R^2 from the firm-level news regression,
2. mean percentile rank of news-day volatility,
3. mean news-to-non-news volatility ratio,
4. median news-to-non-news volatility ratio.

The composite measure is the average of these standardized components.

Interpretation. This reduces dependence on any one way of quantifying the importance of media news.

3.9 Stochastic Search Variable Selection (SSVS)

The paper uses Bayesian Stochastic Search Variable Selection to compare many possible country-level explanations in a small cross-section of countries. The model is

$$Y = X\beta + \varepsilon, \quad \varepsilon \sim N(0, \sigma^2 I), \quad (13)$$

$$\sigma^2 \sim \frac{\nu\lambda}{\chi_\nu^2}, \quad (14)$$

$$\beta_i \mid \gamma_i \sim \begin{cases} N(0, v_1), & \gamma_i = 1, \\ N(0, v_0), & \gamma_i = 0, \end{cases} \quad (15)$$

$$f(\gamma) \sim \prod_i p_i^{\gamma_i} (1 - p_i)^{1-\gamma_i}. \quad (16)$$

Here $\gamma_i = 1$ means variable i belongs in the important-variable set. The paper reports posterior probabilities that each variable is important.

Interpretation. SSVS is useful because the authors have many plausible country variables but few countries. It penalizes models for trying many possible explanations and focuses on variables that are economically and statistically important.

3.10 Takeover leakage measure

For mergers, the paper examines buy-and-hold abnormal returns from 55 trading days before the announcement through two days after the announcement. A key object is the fraction of the total run-up that occurs before the official public news:

$$LeakageShare = \frac{BHAR[-55, -2]}{BHAR[-55, 2]}. \quad (17)$$

Interpretation. If low public-news-reaction markets have high pre-announcement merger run-ups, that supports the idea that information leaks into prices before public media announcements.

3.11 Return-reversal tests

The paper defines an extreme return as a daily excess return at least two standard deviations away from the mean excess return over the prior 250 trading days. It then studies buy-and-hold excess returns from day +1 to day +20 after the extreme return day.

Interpretation. If non-news-day extreme returns mainly reflect liquidity shocks, they should reverse. If they reflect informed trading, they should reverse less. The paper therefore asks whether reversals on non-news days are weaker in low-news-reaction countries.

4 Identification and credibility (what makes the estimates persuasive)

4.1 This is not a clean natural experiment

The paper does not use a single shock, instrument, or randomized design. The evidence is mainly cross-country and therefore observational. The credibility of the paper comes from using several complementary designs that all point toward the same mechanism.

4.2 Why earnings announcements are a strong starting point

Earnings announcements are useful because they are common, standardized, and economically meaningful across countries. A comparison based only on arbitrary news stories could be confounded by different types of news being written in different markets. Earnings announcements reduce that problem.

Key point. The paper first establishes that the developed/emerging difference appears for a comparable type of public information event.

4.3 Why date checking matters

The authors devote substantial effort to checking earnings dates against Factiva. This matters because measurement error in event dates would mechanically reduce measured announcement reactions. The paper's method tries to isolate the first public earnings news, not a stale database date.

Credibility contribution. The result is less likely to be an artifact of using noisy international earnings-announcement dates.

4.4 Why firm-level controls matter

The paper checks whether country differences survive controls for size, industry, prior liquidity, prior coverage, and event-date accuracy. This is important because developed markets may have larger firms, different industries, and better trading conditions.

Result. The basic ranking remains similar after these controls, suggesting that country-level information environments matter beyond firm composition.

4.5 Why the general-news tests matter

The earnings results could have been special to accounting news. The authors therefore repeat the analysis using all firm-level media news. Developed markets again show stronger relations between news and returns.

Credibility contribution. The result is not driven only by earnings announcements.

4.6 Why multiple news definitions matter

The paper varies how news is classified:

- pre-trading-day news only,

- all intraday news with local time stamps,
- all articles even without time stamps,
- only major international financial-news sources,
- samples excluding earnings windows.

The broad pattern remains: news explains returns more strongly in developed markets.

4.7 Why SSVS is useful

In cross-country finance, many variables are correlated: GDP per capita, accounting quality, press freedom, investor protection, technology, and market sophistication. Ordinary regressions can be sensitive to which controls are included.

SSVS makes this sensitivity explicit by assigning posterior probabilities to candidate variables. Insider trading and technological development repeatedly emerge as important, while some commonly used country variables do not.

4.8 Why takeover leakage is a powerful auxiliary test

If weak news reactions are caused by public information being learned in advance through legal public channels, then takeover run-ups need not be especially large before the first public merger news. But if private information leakage is important, prices should rise before official takeover announcements.

The paper finds much more pre-announcement run-up in low-news-reaction countries. This is exactly what the insider-trading interpretation predicts.

4.9 Why the reversal test is informative

Prior U.S. evidence shows that reversals are concentrated after large non-news-day moves. The intuition is that non-news-day moves are more likely to reflect liquidity shocks.

The paper shows that non-news-day reversals are smaller in low-news-reaction countries. That is consistent with non-news-day price movement in those countries containing more private information and less pure liquidity trading.

4.10 Main limitations

- **Country-level omitted variables.** Insider trading, technology, disclosure, press freedom, and development are correlated, so the regressions cannot fully isolate causal effects.
- **Survey-based insider trading measure.** The central insider-trading variable is based on executives' perceptions, not direct illegal trades.
- **Media coverage sample selection.** The sample is tilted toward firms that have Factiva coverage and actively traded stocks.
- **News content is not deeply parsed.** The main tests use the presence and number of articles,

Table 1
News summary statistics

	Total firms	Total articles	Avg. % Eng.	Avg. % Bus.	Avg. % WC <250	Emerging	Total firms	Total articles	Avg. % Eng.	Avg. % Bus.	Avg. % WC <250
Developed	20	6,155	94.0	14.4	57.3	1	11,001	54.2	11.9	27.1	27.1
Australia	20	20,400	94.0	14.4	57.3	1	11,001	54.2	11.9	27.1	27.1
Austria	20	20,400	94.0	14.4	57.3	1	11,001	54.2	11.9	27.1	27.1
Belgium	19	13,153	99.7	11.1	42.2	8	2,552	98.2	1.7	40.4	40.4
Cyprus	8	2,178	99.7	11.1	42.2	8	2,552	98.2	1.7	40.4	40.4
Czechia	17	2,086	94.0	15.5	76.9	231	31,255	96.9	3.7	99.9	99.9
Denmark	26	13,198	28.1	20.6	69.5	20	9,444	33.4	28.1	99.5	99.5
Finland	25	14,064	54.9	16.1	66.0	2	705	82.8	21.0	99.2	99.2
France	13	4,985	28.3	15.8	64.9	3	705	60.8	26.3	49.2	49.2
Germany	13	15,878	40.0	12.0	42.9	30	2,795	96.3	14.5	81.0	81.0
Greece	43	5,496	92.9	13.2	45.9	13	1,176	64.2	0.8	65.1	65.1
Hong Kong	188	22,944	92.9	19.6	37.7	28	12,442	66.4	14.4	65.9	65.9
India	27	84,864	18.9	16.5	59.1	138	57,403	96.5	7.3	90.7	90.7
Israel	36	4,684	92.8	8.4	51.2	28	4,949	96.5	12.2	62.2	62.2
Italy	27	84,864	18.9	16.5	59.1	138	57,403	96.5	7.3	90.7	90.7
Japan	39	7,932	86.8	56.1	79.4	8	3,847	62.2	1.2	42.8	42.8
Netherlands	29	21,702	24.7	40.7	65.7	11	13,105	52.6	1.1	58.0	58.0
New Zealand	23	13,119	99.6	8.6	49.1	154	21,202	93.6	13.5	61.9	61.9
Norway	23	12,165	45.4	39.2	65.7	19	8,067	42.3	29.3	58.3	58.3
Portugal	13	12,840	26.6	30.6	60.4	21	1,774	57.2	26.2	73.4	73.4
Singapore	114	23,469	93.0	16.4	61.5	29	2,446	98.4	18.4	33.6	33.6
Spain	25	53,052	21.6	15.4	71.8	22	4,010	47.9	23.0	68.9	68.9
Sweden	30	16,497	35.1	25.4	61.5	48	17,963	33.1	5.0	55.1	55.1
Switzerland	17	13,225	11.2	11.7	59.8	30	1,460	91.5	4.3	65.9	65.9
Taiwan	217	46,700	93.6	49.8	60.1	11	1,354	88.5	7.6	69.8	69.8
U.K.	17	3,495	97.8	26.9	47.9	2	1,032	67.2	2.0	64.2	64.2
U.S.	341	183,749	96.1	20.7	43.5	46	11,898	96.9	12.9	48.9	48.9
						58	10,822	53.1	14.1	79.1	79.1
						5	3,378	28.5	46.2	71.5	71.5
Total	1,342	572,987	62.6	21.8	56.2	1,251	298,614	71.8	14.0	59.1	59.1
Mean	51.6	22,038.0	62.6	21.8	56.2	41.7	9,953.8	71.8	14.0	59.1	59.1
Dev.-Eng.			-9.2	7.8	-2.9						
(t-stat.)			(-1.14)	(2.28)	(-0.83)						

We identify firm news using full-text searches for the firm's name in the headline or lead paragraph of articles in Factset. Firm names are the Datastream expanded name field with common character strings (A, DEAD, CO, SA, etc.) removed. The sample is constructed as discussed in Section 2 and footnote 9. "Avg. % Eng." is the average percentage of articles in a firm year written in English. "Avg. % Bus." is the average percent of articles in a firm year that are published by Dow Jones, Reuters, the Financial Times, or The Wall Street Journal. "Avg. % WC <250" is the average percent of articles in a firm year with less than 250 words. A country is classified as developed if the gross national income per capita is greater than 10,725 U.S. dollars in calendar year 2005, following the World Bank's classification for "high income" countries. All other countries are classified as emerging. OECD membership is from calendar year 2005.

not detailed textual sentiment or topic-specific content.

- **Local language and time-stamp issues.** The authors address these carefully, but international news data inevitably contain measurement challenges.

5 Tables and figures

5.1 Table 1: news summary statistics

Read. The final news sample includes 1,342 developed-market firms and 1,251 emerging-market firms. Developed-market firms have 572,987 articles; emerging-market firms have 298,614. Developed firms average more articles per firm and more news days.

Economic message. Developed markets have more media coverage. But the paper's key contribution is to show that more coverage is also more connected to price discovery.

5.2 Table 2: earnings and merger announcement samples

Table 2
Earnings and merger announcement summary statistics

	Earnings News	Merger News		Earnings News	Merger News
Developed Countries	3,504	2,079	Emerging Countries	466	105
Australia	46	1	Argentina	8	0
Austria	46	1	Brazil	8	0
Belgium	46	1	Canada	8	0
Cyprus	46	1	China	8	0
Czechia	46	1	Colombia	8	0
Denmark	46	1	Croatia	8	0
Finland	46	1	Czech Republic	8	0
France	46	1	Denmark	8	0
Germany	46	1	Finland	8	0
Greece	46	1	France	8	0
Hong Kong	46	1	Germany	8	0
India	46	1	Greece	8	0
Israel	46	1	Hong Kong	8	0
Italy	46	1	India	8	0
Japan	46	1	Israel	8	0
Netherlands	46	1	Italy	8	0
New Zealand	46	1	Japan	8	0
Norway	46	1	Netherlands	8	0
Portugal	46	1	New Zealand	8	0
Singapore	46	1	Norway	8	0
Spain	46	1	Portugal	8	0
Sweden	46	1	Singapore	8	0
Switzerland	46	1	Spain	8	0
Taiwan	46	1	Sweden	8	0
U.K.	46	1	Switzerland	8	0
U.S.	46	1	Taiwan	8	0
			U.K.	8	0
			U.S.	8	0
Total	3,504	2,079	Total	466	105
Average	77	139	Average	40	80

Table 2
Continued

	1-10-05	1-10-10	1-10-15	1-10-20	1-10-25	1-10-30	1-10-35	1-10-40	1-10-45	1-10-50	1-10-55	1-10-60	1-10-65	1-10-70	1-10-75	1-10-80	1-10-85	1-10-90	1-10-95	1-10-00	1-10-05	1-10-10	1-10-15	1-10-20	1-10-25	1-10-30	1-10-35	1-10-40	1-10-45	1-10-50	1-10-55	1-10-60	1-10-65	1-10-70	1-10-75	1-10-80	1-10-85	1-10-90	1-10-95	1-10-00	1-10-05	1-10-10	1-10-15	1-10-20	1-10-25	1-10-30	1-10-35	1-10-40	1-10-45	1-10-50	1-10-55	1-10-60	1-10-65	1-10-70	1-10-75	1-10-80	1-10-85	1-10-90	1-10-95	1-10-00	1-10-05	1-10-10	1-10-15	1-10-20	1-10-25	1-10-30	1-10-35	1-10-40	1-10-45	1-10-50	1-10-55	1-10-60	1-10-65	1-10-70	1-10-75	1-10-80	1-10-85	1-10-90	1-10-95	1-10-00	1-10-05	1-10-10	1-10-15	1-10-20	1-10-25	1-10-30	1-10-35	1-10-40	1-10-45	1-10-50	1-10-55	1-10-60	1-10-65	1-10-70	1-10-75	1-10-80	1-10-85	1-10-90	1-10-95	1-10-00	1-10-05	1-10-10	1-10-15	1-10-20	1-10-25	1-10-30	1-10-35	1-10-40	1-10-45	1-10-50	1-10-55	1-10-60	1-10-65	1-10-70	1-10-75	1-10-80	1-10-85	1-10-90	1-10-95	1-10-00	1-10-05	1-10-10	1-10-15	1-10-20	1-10-25	1-10-30	1-10-35	1-10-40	1-10-45	1-10-50	1-10-55	1-10-60	1-10-65	1-10-70	1-10-75	1-10-80	1-10-85	1-10-90	1-10-95	1-10-00	1-10-05	1-10-10	1-10-15	1-10-20	1-10-25	1-10-30	1-10-35	1-10-40	1-10-45	1-10-50	1-10-55	1-10-60	1-10-65	1-10-70	1-10-75	1-10-80	1-10-85	1-10-90	1-10-95	1-10-00	1-10-05	1-10-10	1-10-15	1-10-20	1-10-25	1-10-30	1-10-35	1-10-40	1-10-45	1-10-50	1-10-55	1-10-60	1-10-65	1-10-70	1-10-75	1-10-80	1-10-85	1-10-90	1-10-95	1-10-00	1-10-05	1-10-10	1-10-15	1-10-20	1-10-25	1-10-30	1-10-35	1-10-40	1-10-45	1-10-50	1-10-55	1-10-60	1-10-65	1-10-70	1-10-75	1-10-80	1-10-85	1-10-90	1-10-95	1-10-00	1-10-05	1-10-10	1-10-15	1-10-20	1-10-25	1-10-30	1-10-35	1-10-40	1-10-45	1-10-50	1-10-55	1-10-60	1-10-65	1-10-70	1-10-75	1-10-80	1-10-85	1-10-90	1-10-95	1-10-00	1-10-05	1-10-10	1-10-15	1-10-20	1-10-25	1-10-30	1-10-35	1-10-40	1-10-45	1-10-50	1-10-55	1-10-60	1-10-65	1-10-70	1-10-75	1-10-80	1-10-85	1-10-90	1-10-95	1-10-00	1-10-05	1-10-10	1-10-15	1-10-20	1-10-25	1-10-30	1-10-35	1-10-40	1-10-45	1-10-50	1-10-55	1-10-60	1-10-65	1-10-70	1-10-75	1-10-80	1-10-85	1-10-90	1-10-95	1-10-00	1-10-05	1-10-10	1-10-15	1-10-20	1-10-25	1-10-30	1-10-35	1-10-40	1-10-45	1-10-50	1-10-55	1-10-60	1-10-65	1-10-70	1-10-75	1-10-80	1-10-85	1-10-90	1-10-95	1-10-00	1-10-05	1-10-10	1-10-15	1-10-20	1-10-25	1-10-30	1-10-35	1-10-40	1-10-45	1-10-50	1-10-55	1-10-60	1-10-65	1-10-70	1-10-75	1-10-80	1-10-85	1-10-90	1-10-95	1-10-00	1-10-05	1-10-10	1-10-15	1-10-20	1-10-25	1-10-30	1-10-35	1-10-40	1-10-45	1-10-50	1-10-55	1-10-60	1-10-65	1-10-70	1-10-75	1-10-80	1-10-85	1-10-90	1-10-95	1-10-00	1-10-05	1-10-10	1-10-15	1-10-20	1-10-25	1-10-30	1-10-35	1-10-40	1-10-45	1-10-50	1-10-55	1-10-60	1-10-65	1-10-70	1-10-75	1-10-80	1-10-85	1-10-90	1-10-95	1-10-00	1-10-05	1-10-10	1-10-15	1-10-20	1-10-25	1-10-30	1-10-35	1-10-40	1-10-45	1-10-50	1-10-55	1-10-60	1-10-65	1-10-70	1-10-75	1-10-80	1-10-85	1-10-90	1-10-95	1-10-00	1-10-05	1-10-10	1-10-15	1-10-20	1-10-25	1-10-30	1-10-35	1-10-40	1-10-45	1-10-50	1-10-55	1-10-60	1-10-65	1-10-70	1-10-75	1-10-80	1-10-85	1-10-90	1-10-95	1-10-00	1-10-05	1-10-10	1-10-15	1-10-20	1-10-25	1-10-30	1-10-35	1-10-40	1-10-45	1-10-50	1-10-55	1-10-60	1-10-65	1-10-70	1-10-75	1-10-80	1-10-85	1-10-90	1-10-95	1-10-00	1-10-05	1-10-10	1-10-15	1-10-20	1-10-25	1-10-30	1-10-35	1-10-40	1-10-45	1-10-50	1-10-55	1-10-60	1-10-65	1-10-70	1-10-75	1-10-80	1-10-85	1-10-90	1-10-95	1-10-00	1-10-05	1-10-10	1-10-15	1-10-20	1-10-25	1-10-30	1-10-35	1-10-40	1-10-45	1-10-50	1-10-55	1-10-60	1-10-65	1-10-70	1-10-75	1-10-80	1-10-85	1-10-90	1-10-95	1-10-00	1-10-05	1-10-10	1-10-15	1-10-20	1-10-25	1-10-30	1-10-35	1-10-40	1-10-45	1-10-50	1-10-55	1-10-60	1-10-65	1-10-70	1-10-75	1-10-80	1-10-85	1-10-90	1-10-95	1-10-00	1-10-05	1-10-10	1-10-15	1-10-20	1-10-25	1-10-30	1-10-35	1-10-40	1-10-45	1-10-50	1-10-55	1-10-60	1-10-65	1-10-70	1-10-75	1-10-80	1-10-85	1-10-90	1-10-95	1-10-00	1-10-05	1-10-10	1-10-15	1-10-20	1-10-25	1-10-30	1-10-35	1-10-40	1-10-45	1-10-50	1-10-55	1-10-60	1-10-65	1-10-70	1-10-75	1-10-80	1-10-85	1-10-90	1-10-95	1-10-00	1-10-05	1-10-10	1-10-15	1-10-20	1-10-25	1-10-30	1-10-35	1-10-40	1-10-45	1-10-50	1-10-55	1-10-60	1-10-65	1-10-70	1-10-75	1-10-80	1-10-85	1-10-90	1-10-95	1-10-00	1-10-05	1-10-10	1-10-15	1-10-20	1-10-25	1-10-30	1-10-35	1-10-40	1-10-45	1-10-50	1-10-55	1-10-
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5.3 Figure 1: earnings-announcement reactions

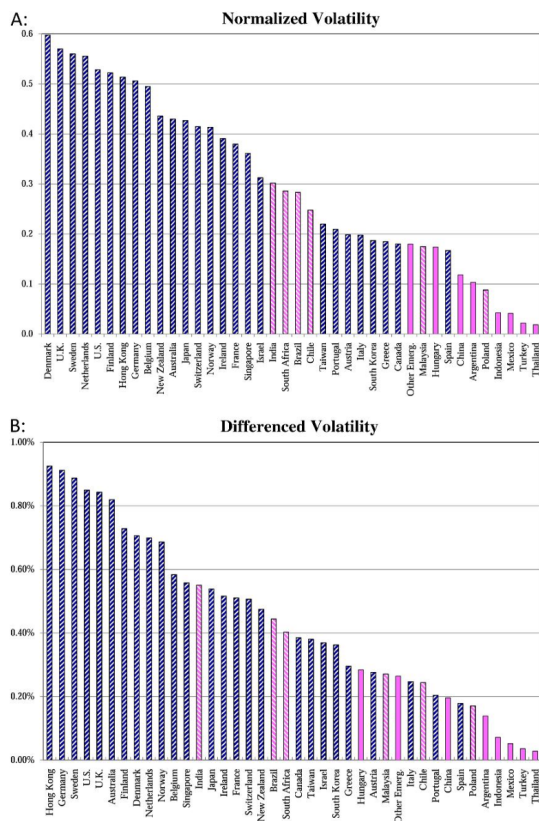


Figure 1
(Continued)

Read. Denmark, the United Kingdom, Sweden, the Netherlands, the United States, Finland, Hong Kong, and Germany have earnings reactions above 0.50, meaning event-window volatility is more than 50% above normal. Thailand, Turkey, Mexico, Indonesia, Poland, Argentina, and China have reactions below 0.15.

Economic message. Public earnings news causes large stock-price movements in many developed markets but little incremental movement in many emerging markets. The official announcement is much less of a surprise in low-reaction markets.

5.4 Table 3: earnings event reactions

Table 3
Earnings event reactions

Panel A: Developed Markets

Country	Normalized Volatility			Differenced Volatility (%)			Residual Differenced Volatility		
	S	M	L	MB-AJ	DM-AJ	DM-AJ	MB-AJ	DM-AJ	DM-AJ
Australia	0.56	0.47	0.37	0.43	0.37	0.32	0.38	0.32	0.38
Canada	0.56	0.47	0.37	0.43	0.37	0.32	0.38	0.32	0.38
Denmark	0.56	0.47	0.37	0.43	0.37	0.32	0.38	0.32	0.38
France	0.56	0.47	0.37	0.43	0.37	0.32	0.38	0.32	0.38
Germany	0.56	0.47	0.37	0.43	0.37	0.32	0.38	0.32	0.38
Hong Kong	0.56	0.47	0.37	0.43	0.37	0.32	0.38	0.32	0.38
Japan	0.56	0.47	0.37	0.43	0.37	0.32	0.38	0.32	0.38
Netherlands	0.56	0.47	0.37	0.43	0.37	0.32	0.38	0.32	0.38
Sweden	0.56	0.47	0.37	0.43	0.37	0.32	0.38	0.32	0.38
Switzerland	0.56	0.47	0.37	0.43	0.37	0.32	0.38	0.32	0.38
U.S.	0.56	0.47	0.37	0.43	0.37	0.32	0.38	0.32	0.38
Other Emerg.	0.56	0.47	0.37	0.43	0.37	0.32	0.38	0.32	0.38
Mean	0.56	0.47	0.37	0.43	0.37	0.32	0.38	0.32	0.38
Std. Dev.	0.10	0.08	0.06	0.08	0.06	0.04	0.08	0.06	0.04

Table 3
Earnings event reactions

Panel B: Emerging Markets

Country	Normalized Volatility			Differenced Volatility (%)			Residual Differenced Volatility		
	S	M	L	MB-AJ	DM-AJ	DM-AJ	MB-AJ	DM-AJ	DM-AJ
Argentina	0.11	0.08	0.06	0.08	0.06	0.04	0.08	0.06	0.04
China	0.11	0.08	0.06	0.08	0.06	0.04	0.08	0.06	0.04
India	0.11	0.08	0.06	0.08	0.06	0.04	0.08	0.06	0.04
Indonesia	0.11	0.08	0.06	0.08	0.06	0.04	0.08	0.06	0.04
Japan	0.11	0.08	0.06	0.08	0.06	0.04	0.08	0.06	0.04
Malaysia	0.11	0.08	0.06	0.08	0.06	0.04	0.08	0.06	0.04
Other Emerg.	0.11	0.08	0.06	0.08	0.06	0.04	0.08	0.06	0.04
Mean	0.11	0.08	0.06	0.08	0.06	0.04	0.08	0.06	0.04
Std. Dev.	0.02	0.01	0.01	0.02	0.01	0.01	0.02	0.01	0.01

Table 3
Earnings event reactions

Panel C: Emerging Markets

Country	Normalized Volatility			Differenced Volatility (%)			Residual Differenced Volatility		
	S	M	L	MB-AJ	DM-AJ	DM-AJ	MB-AJ	DM-AJ	DM-AJ
Argentina	0.11	0.08	0.06	0.08	0.06	0.04	0.08	0.06	0.04
China	0.11	0.08	0.06	0.08	0.06	0.04	0.08	0.06	0.04
India	0.11	0.08	0.06	0.08	0.06	0.04	0.08	0.06	0.04
Indonesia	0.11	0.08	0.06	0.08	0.06	0.04	0.08	0.06	0.04
Japan	0.11	0.08	0.06	0.08	0.06	0.04	0.08	0.06	0.04
Malaysia	0.11	0.08	0.06	0.08	0.06	0.04	0.08	0.06	0.04
Other Emerg.	0.11	0.08	0.06	0.08	0.06	0.04	0.08	0.06	0.04
Mean	0.11	0.08	0.06	0.08	0.06	0.04	0.08	0.06	0.04
Std. Dev.	0.02	0.01	0.01	0.02	0.01	0.01	0.02	0.01	0.01

Read. Developed markets average 0.38 normalized volatility; emerging markets average 0.15. Developed markets also have 0.56 percentage points of extra absolute abnormal return on earnings

days, compared with 0.23 percentage points in emerging markets. The developed-emerging difference is significant and not driven by market-adjustment method.

Economic message. Earnings announcements are about two and a half times more important in developed markets when measured relative to normal volatility.

5.5 Figure 2 and Table 4: news regression fit

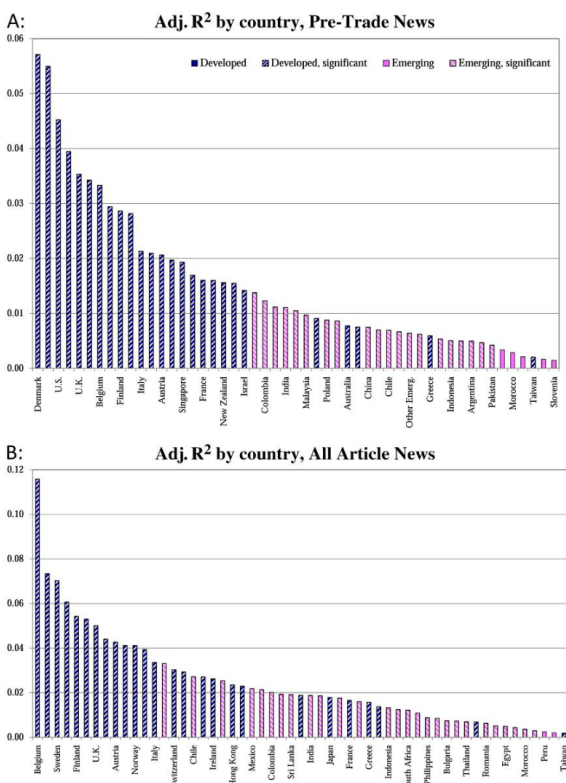


Figure 2
(Continued)

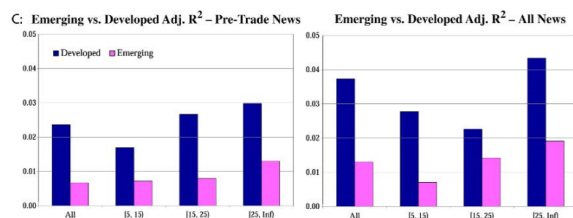


Figure 2
Adjusted R^2 from regressions of returns on news
This figure plots average adjusted R^2 from firm-level regressions of returns on news variables. Details of the calculation of the average adjusted R^2 are discussed in Table 4 and Section 4.1. In Panel A and the left panel of Panel C, the charts include only "Pre-Trade News" articles with time stamps that occurred between the previous day's market close and the current day's market open. In Panel B and the right panel of Panel C, we count all articles as news even if there is no time stamp as in Panel C of Table 4. Emerging markets with fewer than 20 firm years are grouped together in the "Other Emerg." category. Panel C presents the developed- and emerging-market average of the country-level average Adj. R^2 measures. Averages are also presented for firms in three "news bins": 5 to 15 news days in a year, 15 to 25, and 25 and more. Before averaging across countries, we delete a news-bin observation if there are fewer than five firm-year observations in the bin. In Panels A and B, stripes indicate that the Adj. R^2 is significantly higher, using a two-sided bootstrapped t -test with $\alpha=0.05$.

Read. Firm-level regressions of $\ln(1 + |AR|)$ on news-day indicators and article counts have much higher adjusted R^2 in developed markets. Developed markets average about 0.024 using pre-trade news; emerging markets average about 0.007. News improves regression fit for about 40% of developed-market firms and 18% of emerging-market firms.

Economic message. Media news is more useful for explaining idiosyncratic stock volatility in developed markets. This remains true across news-frequency bins and after excluding earnings windows.

5.6 Figure 3: pre-trade news-day versus non-news-day volatility

Read. News-day volatility tends to rank above non-news-day volatility in developed markets. Some emerging markets have large mean news-day ratios, but the median patterns show that the typical emerging-market news day often looks similar to a non-news day.

Economic message. Emerging markets sometimes have large public-news events, but the typical

Table 4
News regression fit (adj. R²)

Country	Firm-Year Count	Panel A1: Pre-Trade Day News Only			
		Adj. R ² - All	News Improves Fit (% of Firms)	Adj. R ² [5,15]	Adj. R ² [15,25]
Australia	44	0.004*	22.7%	0.005*	0.017*
Austria	66	0.011*	42.4	0.002	0.022*
Belgium	70	0.013*	60.0	0.018*	0.042*
Canada	20	0.008*	43	0.026*	0.011*
Cyprus	21	0.009*	33.3	–	–
Denmark	61	0.027*	62.3	0.020*	0.065*
Finland	69	0.029*	43.5	0.039*	0.034*
France	45	0.016*	37.8	0.016*	0.020*
Germany	35	0.016*	40.0	0.007	0.044*
Greece	57	0.008*	24.6	0.015*	–
Hong Kong	474	0.015*	27.4	0.021*	0.008*
Ireland	36	0.020*	33.3	0.017*	0.035*
Israel	33	0.014*	30.2	0.005	0.040*
Italy	80	0.021*	35.0	0.003	0.025*
Japan	159	0.020*	30.3	0.032*	0.030*
Netherlands	71	0.030*	46.5	0.029*	0.040*
New Zealand	70	0.016*	32.9	0.014*	0.027*
Norway	71	0.028*	45.1	0.001	0.029*
Portugal	42	0.014*	45.2	0.010*	0.005
Singapore	242	0.019*	36.8	0.019*	0.035*
Spain	86	0.019*	33.7	0.014*	0.001
Sweden	81	0.005*	22.1	0.006*	0.007*
Switzerland	52	0.021*	40.4	0.008*	0.014*
Taiwan	864	0.002*	4.9	0.001	0.004*
U.K.	37	0.035*	59.5	0.033*	0.031*
U.S.	873	0.045*	53.8	0.018*	0.040*

(continued)

Table 4
Continued

Country	Country Count	Panel A1: Pre-Trade Day News Only (continued)			
		Adj. R ² - All	News Improves Fit (% of Firms)	Adj. R ² [5,15]	Adj. R ² [25,Inf]
Developed	26	0.024	40.0	0.017	0.030
Emerging	25	0.017	17.9	0.007	0.008
Difference	(5.45)	0.007	22.1	0.010	0.017
(t-stat.)		(1.36)	(3.20)	(5.70)	(4.64)
Panel A2: Pre-Trade Day News: Sample with Fictitious confirmed earnings announcements					
Developed	18	0.045	50.7	0.045	0.043
Emerging	13	0.012	25.0	0.010	0.015
Difference	(4.62)	0.032	25.7	0.035	0.028
(t-stat.)		(4.62)	(5.02)	(4.40)	(2.43)
Panel A3: Pre-Trade Day News: Excluding -1 to +2 around earnings announcements					
Developed	18	0.041	47.8	0.041	0.038
Emerging	13	0.012	24.6	0.009	0.014
Difference	(4.62)	0.028	23.2	0.032	0.024
(t-stat.)		(4.62)	(4.43)	(4.40)	(3.03)
Panel B: Local Time Stamped News					
Developed	26	0.039	43.0	0.025	0.041
Emerging	25	0.010	19.0	0.010	0.017
Difference	(6.39)	0.029	24.0	0.015	0.024
(t-stat.)		(6.39)	(5.71)	(5.71)	(4.67)
Panel C: All Available Articles (not timezone adjusted)					
Developed	26	0.037	44.3	0.028	0.023
Emerging	29	0.013	24.0	0.007	0.014
Difference	(4.91)	0.024	20.3	0.021	0.009
(t-stat.)		(4.91)	(5.07)	(2.53)	(1.44)

(continued)

Table 4
Continued

Country	Firm-Year Count	Panel A1: Pre-Trade Day News Only			
		Adj. R ² - All	News Improves Fit (% of Firms)	Adj. R ² [5,15]	Adj. R ² [25,Inf]
Argentina	136	0.005*	16.9	0.006*	0.004*
Brazil	49	0.007*	16.3	0.002	0.003*
Chile	77	0.007*	42.4	0.009*	0.011*
China	418	0.008*	23.2	0.009*	0.005*
Colombia	43	0.012*	25.6	0.005	0.028*
Egypt	86	0.002	4.7	–0.001	0.012*
India	441	0.010*	26.3	0.016*	0.017*
Indonesia	606*	0.005*	11.5	–0.001	0.007
Kenya	100	0.008*	16.0	0.008*	0.006*
Lavos	30	0.003	20.0	0.000	0.000
Latvian	42	0.006*	14.5	0.008*	0.005
Malaysia	432	0.010*	21.3	0.018*	0.011*
Mexico	52	0.014*	25.0	0.014*	0.008*
Morocco	43	0.003	11.6	0.005	0.003
Other Emerging	42	0.006*	14.5	0.005	0.010*
Pakistan	127	0.004*	17.3	0.004*	0.005
Peru	41	0.005*	12.2	0.003	0.008
Philippines	69	0.002	7.2	0.003	0.005
Poland	133	0.009*	20.3	0.007*	0.003
Romania	20	0.005*	20.0	–	–
Slovenia	25	0.001	4.0	0.003	–
South Africa	145	0.007*	24.1	0.012*	0.007*
Thailand	102	0.007*	23.5	0.011*	0.005
Turkey	167	0.011*	26.3	0.011*	0.010*

(continued)

Table 4
Continued

Country	Country Count	Panel D: Dow Jones, Reuters, Financial Times, and The Wall Street Journal, GMT Date Stamp			
		Adj. R ² - All	News Improves Fit (% of Firms)	Adj. R ² [5,15]	Adj. R ² [25,Inf]
Developed	25	0.041	43.0	0.030	0.049
Emerging	23	0.014	23.2	0.015	0.014
Difference	(5.45)	0.027	19.7	0.015	0.035
(t-stat.)		(5.69)	(5.54)	(4.75)	(4.38)

This table shows average adjusted R² from firm-level regressions of returns on news variables derived from the articles summarized in Table 1. The adjusted R² comes from the following regression run each firm year:

$$\ln(1 + |AR_{i,t}|) = \alpha_i + \beta_{NewsDay,t} NewsDay_{i,t-1} + \beta_{ArCCount,t} ArCCount_{i,t-1} + \epsilon_{i,t}$$

where $|AR_{i,t}|$ is firm i 's abnormal volatility on day t calculated as described in Section 3.1, $NewsDay_{i,t-1}$ is an indicator if there is an article with the firm's name in the headline or lead paragraph on day $t-1$, and $ArCCount_{i,t-1}$ is the number of these articles. Panel A only uses pre-trade news, which means articles that appear from the close of the market on day $t-1$ to the market opens on day t . Consequently, Panel A requires that articles have time stamps we can convert to local times. Panel A1 uses all country-firm-years, Panel A2 presents averages over country-firm-years where we have confirmed earnings data. Panel A3 presents averages over country-firm-years where we exclude earnings data. In Panels B through D, we use contemporaneous news, $NewsDay_{i,t}$ and $ArCCount_{i,t}$. Panel B also requires articles have time stamps, but we assign articles to day t if they are published from the close on day $t-1$ to the close on day t . Panel C uses articles even if we do not have their local time stamps and uses the date published, recorded in Greenwich Mean Time, as the date of the article. Panel D is the same as C, except that we restrict the articles to those from Dow Jones, Reuters, Financial Times, or The Wall Street Journal. The top of Panel A1 reports average Adj. R² for countries, while the bottom of Panel A1 and Panels A2 through D report averages over emerging and developed markets and their differences. Counts for countries in the top of Panel A1 list the number of firm-years included, but in the remainder of Panel A1 and Panels A2 through D, the counts are the number of countries in the cross-section. Adj. R²-All lists the average Adj. R² for all firm-years. "News Improves Fit" is the percentage of firm-years where not including the news-day dummy and article counts results in significantly worse fit based on a goodness-of-fit F -test at $\alpha = 0.05$. Adj. R² [5,15] is the average Adj. R² over firm-years with 5 to 15 days with news articles during the year. Adj. R² [15,25] is the average Adj. R² over firm-years with 15 to 25 news days, and Adj. R² [25,Inf] is the average Adj. R² over firm-years with 25 or more news days. To be included, each news count portfolio must contain at least five firm-years. A firm year is included if there are at least 100 trading days with returns, and 20 firm-years per country. Abnormal returns, $AR_{i,t}$, are Winsorized at the 1st and 99th percentiles. A star on a country-level Adj. R² in Panel A1 indicates that the Adj. R² is significantly higher, using a two-sided bootstrapped t -statistic with $\alpha = 0.05$, than specifications where the same number of news days and article counts are randomly assigned the days in the year. To calculate the country-level bootstrapped t -statistics for each of 500 iterations, we randomize the news days within each firm-year sample, estimate the regression described above, then calculate the t -statistic for the difference between the actual Adj. R² and the bootstrapped Adj. R². Significance is based on the distribution of these country-level t -statistics. Tests for the difference between levels in emerging and developed markets are t -statistics based on bootstrapped standard errors from 1,000 draws with replacement from the sample of country-level average Adj. R²s.

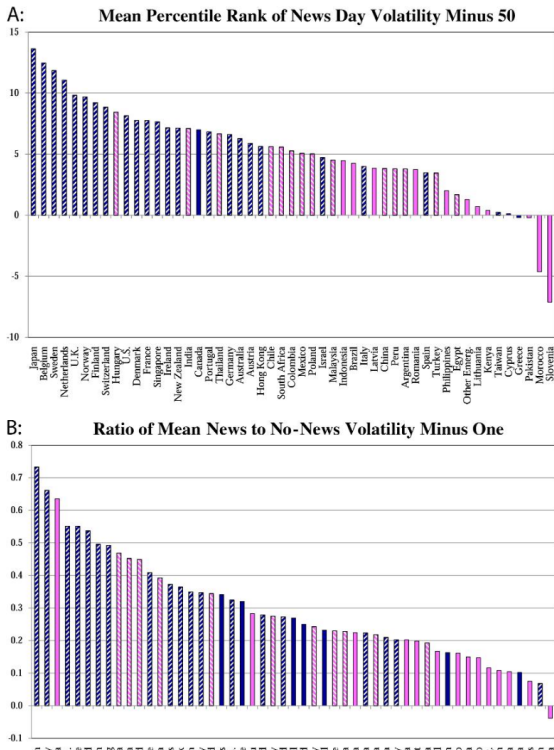


Figure 3
(Continued)

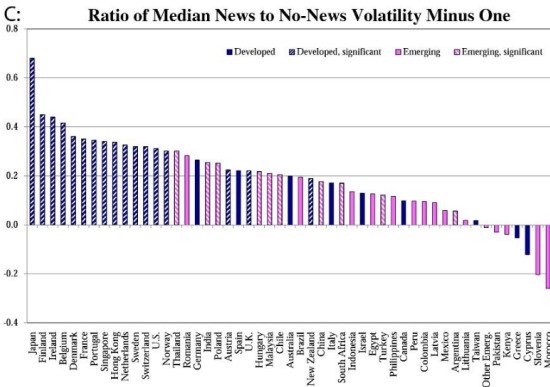


Figure 3
Pre-trade news-day to non-news-day return volatility
This figure plots measures that relate average news-day volatility to non-news-day volatility, where volatility, $|AR_{i,t}|$, is calculated as described in Section 3.1. This figure uses only "Pre-Trade News," as defined in Table 4. Volatility is averaged separately for the firm-news and non-news days to the firm-year level. In Panel A, we rank average firm-year news and non-news volatility from highest to lowest and report the mean rank of the news-day firm-year averages. We divide each rank by the total number of observations to convert the ranks to a percentile and subtract 50 from the percentile for comparability across markets. In Panel B, we average across firm years to get country average news-day and non-news-day volatility, then report the ratio of average news-day to non-news-day volatility minus one. Panel C is the same as B, except using medians. Stripes indicate statistical significance at $\alpha = 0.05$. In Panels A and B, standard errors for the difference between news- and no-news-day volatility are bootstrapped with 100,000 draws with replacement. In Panel C, the tests are Fligner-Policello difference in central tendency tests written in SAS by Paul von Hippel and available at <http://www.sociology.ohio-state.edu/people/ptv/macros/fligner-policello.htm>.

media article is less closely tied to price discovery.

5.7 Table 5: OLS regressions for earnings reactions

Table 5
OLS regressions of ln earnings reactions on country characteristics

Panel A: Regressions with One Variable of Interest

	No Coef.	Adj. R ²	GDP Coef.	GDP	Adj. R ²
Pre-Event Public Information Dissemination In Announcement Order	-0.03 (-0.20)	-0.03	0.05 (0.38)	0.67 (5.31)	0.41
Insider Trading	-0.79 (-7.91)	0.62	-0.67 (-4.52)	0.17 (1.12)	0.62
News Transmission					
Financial Press During Event [-1, 2]	0.37 (2.45)	0.12	0.10 (0.70)	0.62 (4.45)	0.41
News Clustering	0.51 (3.57)	0.24	0.28 (2.15)	0.54 (4.20)	0.47
In-depth Articles During Event [-1, 2]	0.45 (3.04)	0.18	0.18 (1.27)	0.58 (4.18)	0.43
Free Press	0.61 (4.72)	0.36	0.37 (2.77)	0.47 (3.52)	0.51
Tech. Development	0.76 (7.30)	0.57	0.58 (4.35)	0.29 (2.14)	0.61
Accounting Quality					
Accounting Standards	0.80 (8.14)	0.63	0.63 (5.39)	0.77 (2.33)	0.67
Pct. Intl. GAAP	0.40 (2.66)	0.14	0.12 (0.87)	0.62 (4.47)	0.43
Control					
GDP Per Capita	0.66 (5.36)	0.42	- -	- -	- -

(continued)

Read. In one-variable regressions, insider trading, news clustering, in-depth articles, free press, technological development, accounting standards, and international GAAP/IFRS have the expected signs. Insider trading, technological development, and accounting standards have especially high explanatory power and remain important with GDP per capita controls. Announcement order does not explain reactions.

Economic message. The strongest simple explanations for earnings-reaction differences are insider trading, technology/news dissemination, and accounting quality.

5.8 Table 6: SSVS for earnings reactions

Table 6
Regressions of ln earnings reactions on country characteristics with Bayesian model selection

Panel A: Marginal Posterior Probability that the Variable is Important

	0.15	0.5		0.15	0.5
Financial Market Sophistication	0.561	0.097	Pct. Intl. GAAP	0.016	0.072
Accounting Standards	0.288	0.319	Anti-Self-Dealing Index	0.014	0.104
Insider Trading	0.226	0.312	Short Sales Feasible	0.014	0.078
Tech. Development	0.158	0.324	Market Model R ²	0.013	0.066
Financial Disclosure	0.096	0.196	Financial Press During Event	0.013	0.094
Investor Protection Rank	0.074	0.420	Short Sales Legal	0.013	0.075
Country Risk	0.064	0.157	Financial Press Before Event	0.013	0.082
Free Press	0.053	0.199	Insider Trading Enforced	0.012	0.057
In-depth Articles Before Event	0.032	0.204	Disclosure Index	0.012	0.074
In-depth Articles During Event	0.031	0.124	Pct. Days Non-Zero Price Chg.	0.012	0.060
Market Turnover/GDP > 100	0.030	0.143	Director Liability Index	0.012	0.076
U.K. Law	0.029	0.114	Cost to Enforce Contracts	0.011	0.062
GDP per Capita	0.028	0.215	LOF Trading Cost	0.011	0.064
News Clustering	0.020	0.090	Average Lay Firm Size	0.010	0.057
Shareholder Litigation Index	0.018	0.112	Average Firm-Level P/E	0.010	0.059
Investor Protection Index	0.016	0.118			
Only Annual Earnings Ann.					

(continued)

Read. In the Bayesian model-selection exercise for earnings reactions, financial market sophistication, accounting standards, insider trading, and technological development have the highest posterior probabilities under the skeptical prior. Firm-level robustness gives particular support to financial market sophistication and insider trading.

Economic message. The earnings evidence supports a mixed mechanism: private information leakage matters, and the quality of accounting and the broader information environment also matter.

5.9 Table 7: OLS regressions for general news

Read. The dependent variable is a composite general-news reaction measure. With GDP per capita controls, only insider trading and technological development remain strong. Insider trading has a negative coefficient; technological development has a positive coefficient.

Economic message. For news broadly, accounting standards no longer play the central role. The

Table 5
Continued

Panel B: Regressions with Two Variables of Interest

	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]
Insider Trading	-0.43 (-3.04)	-0.65 (-6.04)	-0.50 (-3.78)	-0.77 (-5.98)						
News Clustering				0.04 (0.31)			0.17 (1.30)	0.28 (1.92)	0.39 (2.83)	
Free Press		0.29 (2.67)			0.20 (1.60)			0.35 (3.13)	0.48 (3.31)	
Tech. Development			0.40 (2.97)			0.36 (2.52)		0.61 (5.88)		0.67 (6.46)
Accounting Standards	0.47 (3.30)				0.68 (5.77)	0.52 (3.63)	0.72 (6.54)			
Adj. R ²	0.70	0.67	0.69	0.61	0.65	0.68	0.64	0.66	0.40	0.64

Panel C: Regressions with Three and Four Variables of Interest

	[1]	[2]	[3]	[4]	[5]
Insider Trading	-0.42 (-3.02)	-0.35 (-2.38)	-0.40 (-3.10)		-0.33 (-2.30)
Free Press		0.18 (1.61)		0.25 (2.58)	0.22 (1.91)
Tech. Development			0.23 (1.74)		0.38 (2.74)
Accounting Standards	0.37 (2.41)		0.34 (2.18)	0.28 (2.88)	0.37 (2.36)
Adj. R ²	0.71		0.71	0.73	0.73

This table contains regressions of average natural log of normalized earnings reactions for 39 countries on country characteristics. We consider nine potential explanatory variables that reflect hypotheses about public and private pre-event information dissemination, news transmission, and accounting quality. Variables are described in detail in Appendix Table A1. All variables are standardized. Panel A presents regressions with one explanatory variable. Panel B presents regressions with two explanatory variables using all combinations of variables that are significant in Panel A when GDP Per Capita is used as a control. Panel C presents regressions with three and four explanatory variables using variables that were consistently significant in the Panel B regressions. Parentheses indicate *t*-statistics.

Table 6
Continued

Panel C: Firm-Level Marginal Posterior Probability that Top 10 Variables are Important					
Setting for p	0.15	0.5		0.15	0.5
Financial Market Sophistication	0.120	0.427	Market Turnover/GDP $\times 100$	0.011	0.050
Insider Trading	0.020	0.083	Free Press	0.009	0.045
Investor Protection Index	0.017	0.063	U.K. Law	0.008	0.061
Accounting Standards	0.015	0.062	Financial Press Article During Event	0.008	0.047
Shareholder Lawsuits Index	0.013	0.092	Financial Disclosure	0.008	0.038

This table contains regressions of the natural log of one plus the normalized volatility reported in Table 3 for 39 countries on 33 potential explanatory variables described in Online Table A.1. All variables are standardized by subtracting the cross-country mean and dividing by the standard deviation. We use the Bayesian Stochastic Search Variable Selection methodology of George and McCulloch (1993).

$$y = X\beta + \epsilon, \epsilon \sim N(0, \sigma^2), \quad (1)$$

$$\sigma^2 \sim \nu_0 / j_0^2, \quad (2)$$

$$\beta_j | \tau_j \sim \begin{cases} N(0, \nu_1) & \text{where } \tau_j = 1 \\ N(0, \nu_0) & \text{where } \tau_j = 0 \end{cases}, \quad (3)$$

$$j(\gamma) = \prod_{j=1}^p \lambda_j^{\gamma_j} (1 - p_j)^{1-\gamma_j}, \quad (4)$$

The priors are controlled by the hyperparameters j , ν_1 , ν_0 , ν_1 , and p . One may think of j as an estimate for residual variance and ν as the sample size for the residual variance estimate. We choose the sample variance of the dependent variable as our value for j , and we set ν_1 at 3. The ν_0 and ν_1 hyperparameters are the prior variances of coefficients on unimportant and important independent variables, respectively. We believe that an economically significant variable is one for which a one-standard-deviation change leads to at least a one-tenth standard-deviation change in the dependent variable, which is $|\alpha|(\text{mean}(\text{Normalized Volatility}) - \text{min}(\text{Normalized Volatility}))/10\%$ of Countries). This threshold corresponds to any ν_0 and ν_1 satisfying $0.1^2 = \log(\nu_1/\nu_0)/(\nu_0^{-1} - \nu_1^{-1})$, and we use $\nu_1/\nu_0 = 3.233$. We consider two values for p , the prior probability that an independent variable is important: 0.5 and 0.15. We use the Gibbs sampler to obtain 1,000,000 draws from the posterior distribution $j(\gamma)$ after a 1,000 draw burn-in period. Panel A reports the marginal posterior probability that $\tau_j = 1$ for all independent variables. In Panel B, we identify the four models with the highest posterior probability from each of the two prior specifications and report mean values of the important coefficients from these model draws. The numbers in brackets are 95% credible intervals. We also run the procedure at the firm level, with the dependent variable denominated by size and industry portfolio groups. We use U.S. size tercile breakpoints and Fama and French 48 industries. Panel C reports posterior probabilities for the top 10 variables from the firm-level results.

main story becomes private information leakage plus the quality of news transmission.

5.10 Table 8: SSVS for general news

Table 8
Regressions of the general news reaction measure on country characteristics with Bayesian model selection

Panel A: Marginal Posterior Probability that the Variable is Important					
Setting for p	0.15	0.5		0.15	0.5
Tech. Development	0.635	0.736	Short Sales Feasible	0.021	0.120
Insider Trading	0.369	0.664	Cost to Enforce Contracts	0.019	0.094
In-depth Articles Before Event	0.108	0.556	Anti-Self-Dealing Index	0.017	0.124
Country Risk	0.072	0.255	Director Liability Index	0.017	0.132
Financial Disclosure	0.053	0.440	Investor Protection Rank	0.016	0.121
U.K. Law	0.036	0.231	Short Sales Legal	0.016	0.096
Disclosure Index	0.033	0.207	Free Press	0.016	0.090
Average Firm-Level P/E	0.032	0.165	Only Annual Earnings Ann.	0.014	0.104
Investor Protection Index	0.029	0.176	Pct. Daily Non-Gen. Price Chg.	0.014	0.081
Financial Market Sophistication	0.027	0.156	Market Model R^2	0.014	0.076
Accounting Standards	0.027	0.143	Financial Press Before Event	0.014	0.083
Market Turnover/GDP $\times 100$	0.025	0.207	Insider Trading Enforced	0.014	0.076
Shareholder Lawsuits Index	0.023	0.140	LOF Trading Cost	0.013	0.076
Pct. Intl. GAAP	0.023	0.127	Average Log Firm Size	0.013	0.092
GDP per Capita	0.022	0.131			

Panel B: Best Models							
Setting for $p=0.15$				Setting for $p=0.50$			
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Tech. Development	0.64 [0.35, 0.94]	0.42 [0.04, 0.80]	0.73 [0.43, 1.02]	0.58 [0.24, 0.94]	0.64 [0.35, 0.93]	0.55 [0.17, 0.93]	0.46 [0.11, 0.81]
Insider Trading	-0.63 [-0.93, -0.33]	-0.35 [-0.74, 0.03]	-0.71 [-1.13, -0.31]	-0.33 [-0.57, -0.08]	-0.64 [-1.08, -0.20]	-0.43 [-0.82, -0.09]	-0.34 [-0.61, -0.08]
In-depth Articles Before Event			-0.28 [-0.54, 0.00]				
Financial Disclosure						-0.51 [-0.93, -0.08]	
Pct. Intl. GAAP	0.304	0.155	0.027	0.025	0.008	0.005	0.005

(continued)

Table 7
OLS regressions of ln general news on country characteristics

Panel A: Regressions with One Variable of Interest					
	No controls		GDP control		
	Coef.	Adj. R ²	Coef.	GDP	Adj. R ²
Pre-Event Public Information Dissemination					
In Announcement Order	-0.04 (-0.31)	-0.02	0.00 (0.03)	0.33 (3.11)	0.17
Insider Trading					
Insider Trading	-0.46 (-4.97)	0.39	-0.46 (-3.34)	-0.01 (-0.07)	0.37
News Transmission					
Financial Press During Event [-1, 2]	0.24 (2.04)	0.08	0.10 (0.76)	0.29 (2.41)	0.19
News Clustering	0.29 (2.43)	0.12	0.16 (1.22)	0.27 (2.26)	0.21
In-depth Articles During Event [-1, 2]	0.12 (1.02)	0.00	-0.05 (-0.43)	0.36 (2.96)	0.18
Free Press	0.23 (2.02)	0.08	0.07 (0.58)	0.30 (2.37)	0.18
Tech. Development	0.47 (5.18)	0.41	0.43 (3.60)	0.05 (0.44)	0.40
Accounting Quality					
Accounting Standards	0.36 (3.50)	0.23	0.25 (1.92)	0.18 (1.40)	0.25
Pct. Intl. GAAP	0.21 (1.78)	0.06	0.06 (0.50)	0.31 (2.54)	0.18
Control					
GDP Per Capita	0.33 (3.17)	0.20	-	-	-

Panel B: Regression with Two Variables of Interest

	[1]
Insider Trading	-0.24 (-1.88)
Tech. Development	0.29 (2.20)
Adj. R ²	0.45

This table contains regressions of a country-level measure of market reaction to news for 38 countries on country characteristics. The measure is the average of four standardized measures: the Mean Percentile Rank of News Day Volatility, the Ratio of Mean News to No-News Volatility, the Ratio of Median News to No-News Volatility, and the News Regression Fit (adj. R^2). Construction of the three volatility measures is discussed in Figure 3, and Table 4 discusses the adj. R^2 measure. We consider nine potential explanatory variables which reflect hypotheses about pre-event information dissemination, public and private, news transmission, and accounting quality. Variables are described in detail in Appendix Table A1. All variables are standardized. Panel A presents regressions with one explanatory variable. Panel B presents one regression with the two explanatory variables that are significant in Panel A when GDP Per Capita is used as a control. Parentheses indicate t -statistics.

5.11 Figure 4: merger run-ups

Read. Countries are split into high- and low-earnings-reaction groups. In high-reaction countries, 14% of the total takeover run-up has occurred by two days before the first public merger news. In low-reaction countries, 39% has occurred by then. The difference is statistically significant.

Economic message. Low public-news-reaction countries show nearly three times more pre-announcement merger leakage. This strongly supports the insider-trading interpretation.

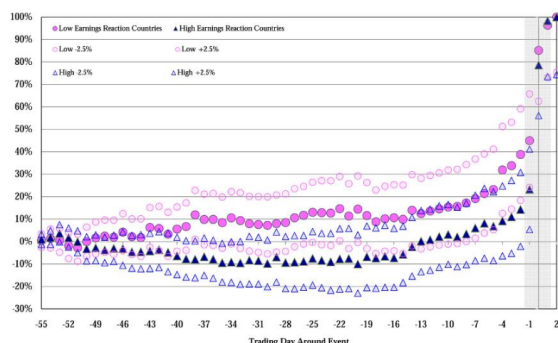


Figure 4

Merger buy-and-hold abnormal returns (BHARs) as a percentage of BHARs for the entire period

This figure shows buy-and-hold abnormal returns (BHARs) for mergers from 55 days before an announcement through two days after an announcement as a percent of the BHAR for the entire [-55, 2] period. The sample of merger announcements is collected from Bloomberg, Mergerstat, and SDC. We restrict the sample to initial bids (no bids for the target in the prior two years) and mergers where the target has at least one article written about it between 60 calendar days and two trading days prior to the announcement and no merger-related articles in the same time frame. A merger event in any of the three sources is considered the same event as one from another source if the bids are for the same target and within two years of each other. We take the earliest announcement date for each event from the union of all three sources. Events are divided according to whether the firm's home country has an average earnings announcement reaction above or below the median for all countries with at least 20 earnings announcements and SUEs as calculated in Figure 1, Panel A. There are 293 events in the high-reaction group and 278 events in the low-reaction group. Abnormal returns are market adjusted, which means they are the buy-and-hold return for a stock minus the buy-and-hold return for the market. In order for an event to be included, the stock must have 50% of trading days with price changes in the calendar year prior to the event. The shaded region in each panel marks the [-1, 2] event window. 95% confidence intervals for the BHARs are marked with smaller, lighter circles or triangles.

Table 9

Buy-and-hold excess returns from days +1 to +20 following extreme return events

Panel A: Buy-and-Hold Excess Returns					
Extreme Return with	No News		News		
	Neg. Ret (-)	Pos. Ret (+)	Neg. Ret (-)	Pos. Ret (+)	
High Reaction	1.31*	-0.91*	-0.33	0.35	
Low Reaction	0.55*	-0.22	-0.27	0.82*	
Difference	0.76	-0.68	-0.06	-0.48	
t-test	2.02	-2.22	-0.12	-1.07	
p-value	(0.04)	(0.03)	(0.90)	(0.28)	

Panel B: Buy-and-Hold Excess Returns as a Percentage of Day Zero Returns					
Extreme Return with	No News		News		
	Neg. Ret (-)	Pos. Ret (+)	Neg. Ret (-)	Pos. Ret (+)	
High Reaction	-19.14*	-11.10*	4.19	4.07	
Low Reaction	-8.85*	-3.18	4.49	12.11*	
Difference	-10.32	-8.16	-0.31	-8.14	
t-test	-1.80	-2.00	-0.04	-1.15	
p-value	(0.07)	(0.05)	(0.97)	(0.18)	

This table presents buy-and-hold excess returns (return minus the value-weighted country portfolio) following days with extreme returns. We assign days to bins depending on whether the extreme return is positive or negative, whether the stock is from a high- or low-reaction country, and whether there is or is not news for the stock on the extreme return day. An extreme return day is a day with excess returns at least two standard deviations away from the mean compared to excess returns over the previous 250 trading days. High and low-reaction country classification is defined as in Figure 4. We require all stocks to have at least one news article in the prior 60 trading days to ensure that these are firms covered by our news source and no other extreme returns in the 30 days prior to the observed extreme return event to avoid clustering of the events. Panel A presents the raw day +1 to +20 buy-and-hold excess returns following days with extreme returns in percent. Panel B presents the average portfolio day +1 to +20 buy-and-hold excess returns as a percent of the average portfolio day zero returns. The difference in means tests use a pooled/unpooled t-test where the null of equal averages between emerging and developed markets is tested. A pooled t-test is used when a folded F-test indicates that sample variances are insignificantly different at the five-percent significance level; otherwise, an unpooled t-test is used. * indicates significance at $\alpha = 0.05$.

5.12 Table 9: reversals after extreme returns

Read. In high-reaction markets, extreme non-news-day returns reverse strongly. For example, large negative non-news-day returns are followed by a 1.31% reversal. In low-reaction markets, the corresponding reversal is only 0.55%. News-day reversals are much weaker or absent.

Economic message. In markets where public news is weakly related to prices, non-news-day price moves appear to contain more information and less liquidity noise.

5.13 Appendix Table A1: variable definitions

Read. The appendix documents the construction of the country variables: insider trading, insider-trading enforcement, free press, technological development, accounting standards, disclosure, international GAAP/IFRS, liquidity, market development, governance, and other controls.

Economic message. The empirical interpretation depends heavily on mapping country variables to economic hypotheses. The appendix clarifies why each variable is included and how it should be signed.

Appendix Table A1
Variable description
Variable

Variable	Description
	Pre-Event Public Information Dissemination
<i>In Announcement Order</i>	All U.S. firms and all firms in sample are sorted in to industry according to Fama's Primary SIC. All firms from Datastream are matched on ISIN and SEDOL. All firms from CRSP are matched on CUSIP. Each annual earnings announcement date is compared to all annual earnings announcement dates from firms in the same Primary SIC within the prior 120 calendar days. All announcements are ranked from first to last and the Announcement Order is this rank. We take the log of the order and average to the country level in cross-country regressions.
	Insider Trading
<i>Insider Trading</i>	The average of the 1999, 2000, and 2002-2003 GCR responses. The question asks if "Insider trading in your country's stock markets is (1=pervasive, 7=extremely rare)." We reverse this variable so, extremely rare is 1 and pervasive is 7.
<i>Insider Trading Enforced</i>	Dummy variable that equals one if insider trading laws exist and are enforced as of 2008 [from Bhattacharya and Daouk (2002) and updated by authors in 2008].
	News Transmission
<i>Financial Press Before Event</i>	Log of one plus the count of articles about the firm in <i>The Wall Street Journal</i> , <i>Financial Times</i> , Dow Jones, or Reuters in the 55 to 2 days before the event.
<i>Financial Press During Event</i>	Log of one plus the count of articles about the firm in <i>The Wall Street Journal</i> , <i>Financial Times</i> , Dow Jones, or Reuters in the -1 to 2 days around the event.
<i>In-depth Articles Before Event</i>	The count of articles about the firm in the window 55 to 2 days before the event with more than 500 words per firm mentioned.
<i>In-depth Articles During Event</i>	The count of articles about the firm in the window -1 to 2 days around the event with more than 500 words per firm mentioned.
<i>News Clustering</i>	The log ratio of one plus the number of articles per day in the window -1 to 2 days around the event to one plus the number of articles per day in the window 55 to 2 days before the event.
<i>Free Press</i>	<i>Free Press</i> is from the 2003, 2004, and 2006 GCR and asks if "the media can publish/broadcast stories of their choosing without fear of censorship or retaliation."
<i>Tech. Development</i>	<i>Technological Development</i> is the average of three variables: The Technological Availability measure from 2007, which asks, "To what extent are the latest technologies available in your country? (1 = not available; 7 = widely available)"; the Technological Readiness measure from 2006, which asks, "What is your country's position in technology relative to world leaders? (1 = Behind; 7 = Ahead)"; and Technological Sophistication from the 2003 GCR, which asks, "Your country's position in technology (1=generally lags behind most countries, 7= is among the world's leaders)".
	Accounting Quality
<i>Accounting Standards</i>	Average response to question about the strength of accounting standards. From the 2003, 2004, 2006, and 2007 Global Competitiveness Report, where 7 is high/strongly agree and 1 is low/strongly disagree on these surveys.
<i>Financial Disclosure</i>	<i>Financial Disclosure</i> is from the 2005-2006 GCR and reports "The level of financial disclosure required is extensive and detailed."
<i>Pct. Intl. GAAP</i>	Percent of Firms Following International Accounting Standards (IFRS) or U.S. GAAP is the percent of firms following those standards in the sample of firms included in the country-level averages (as the dependent variable); U.S. firms are assumed to follow GAAP and Worldscope is the source for the rest of the world.
<i>Only Annual Earn. Ann.</i>	The fraction of firms in the sample that have only annual earnings announcements.

(continued)

Appendix Table A1
Continued
Variable

Variable	Description
<i>Firm Follows GAAP or IFRS</i>	An indicator that is one if the firm follows International Accounting Standards (IFRS) or U.S. GAAP. Worldscope is the source for all foreign firms, and U.S. firms are assumed to follow GAAP.
	Economic & Financial Development
<i>GDP per Capita</i>	$\ln(\text{GDP per Capita})$ in 2000 constant dollars is from World Bank's Financial Structure Dataset developed by Beck, Demirgüç-Kunt, and Levine (2000) and are annual observations averaged over 2003 to 2007.
<i>Market Turnover/GDP x 100</i>	<i>Market Turnover/GDP</i> in 2000 constant dollars is from World Bank's Financial Structure Dataset developed by Beck, Demirgüç-Kunt, and Levine (2000) and are annual observations averaged over 2003 to 2007.
<i>Financial Market Sophistication</i>	The average measure of <i>Financial Market Sophistication</i> from the 2003 through 2007 Global Competitiveness Report and asks if "The level of sophistication of financial markets is higher than international norms."
<i>Ln Firm's Prior Dec. USD MV</i>	Natural log of the firm's prior December market capitalization in U.S. dollars.
	Regulatory Environment
<i>Short Sales Legal</i>	Dummy variable set to 1 if short sales are not against the law (Charoenrook and Daouk 2005).
<i>Short Sales Feasible</i>	Dummy variable set to 1 if short sales are actually used in the market (Charoenrook and Daouk 2005).
<i>UK Law</i>	Dummy variable for whether the legal system in a country is based on common law.
<i>Cost to Enforce Contracts</i>	Costs to enforce contracts as a percentage of debt. Average of the annual measures from 2005 through 2008 from Doingbusiness.org.
	Governance
<i>Investor Protection Rank</i>	Average of the annual measures from 2003 through 2008 from Doingbusiness.org.
<i>Investor Protection Index</i>	Average of the annual measures from 2005 through 2008 from Doingbusiness.org.
<i>Anti-Self-Dealing Index</i>	From Djankov et al. (2008).
<i>Shareholder Lawsuits Index</i>	Average of the annual measures from 2005 through 2008 from Doingbusiness.org.
<i>Director Liability Index</i>	Average of the annual measures from 2005 through 2008 from Doingbusiness.org.
<i>Disclosure Index</i>	Average of the annual measures from 2005 through 2008 from Doingbusiness.org.
	Trading Costs
<i>LOT Trading Cost</i>	Computed following Lesmond et al. (1999). We use the prior year average over firms included in the sample from 2003 through 2008.
<i>Pct. Days Zero Price Chg.</i>	Alternate liquidity measure that is percent of days with zero price changes in the prior year for the firms in the sample from 2003 through 2008.
	Characteristics of Equity Markets
<i>Average Log Firm Size</i>	Natural log of the prior December firm size averaged over all firms in our sample.
<i>Average Firm-Level P/E</i>	Average P/E ratio for firms in sample. P/E data are from Compustat for the U.S. and Datastream and World scope for the rest of the world. Values are from the previous calendar year.
<i>Market Model R²</i>	Natural log of ($R^2/(1-R^2)$) where R^2 is the SST weighted average R^2 of simple market model regressions including the local and the U.S. market for each stock in our sample for each year from 2003 through 2008 (through 2007 in the U.S.) and averaged over all years.
<i>Country Risk</i>	Average over the period 2003-2008 of the country risk index published by Euromoney. Higher values indicate lower risk.

6 Short summary

Griffin, Hirschey, and Kelly study whether the financial media is important for stock-price discovery across global markets. They assemble a large Factiva sample of firm news articles in 56 markets and compare how much stock prices move on public-news days relative to non-news days.

The paper first studies earnings announcements because they are comparable across firms and countries. Developed markets show much larger earnings-announcement reactions than emerging markets: normalized volatility is 0.38 in developed markets and 0.15 in emerging markets. The same pattern appears in general news regressions: media news explains much more idiosyncratic return volatility in developed markets.

The authors evaluate several explanations. The strongest evidence points to insider trading: where insider trading is viewed as more prevalent, public announcements move prices less. Technological development also matters, likely because better technology spreads news faster and to more investors. Accounting standards matter for earnings announcements but not for general news.

Two auxiliary tests strengthen the interpretation. Low-news-reaction countries have much more pre-announcement takeover run-up, consistent with private information leakage. They also have smaller reversals after extreme non-news-day returns, consistent with non-news-day trading being more informed and less liquidity-driven.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that public financial news is not equally informative across global markets. In developed markets, stock prices move strongly when firms receive public media coverage or announce earnings. In many emerging markets, official public news releases generate much smaller incremental price movements.

7.2 Economic intuition

A public announcement moves prices only to the extent that it is a surprise to marginal investors. If information has already leaked through insider trading, then the announcement is less surprising. If technology and news dissemination are weak, the public media channel reaches investors less effectively. If accounting standards are weak, earnings announcements contain less reliable valuation information.

7.3 Takeaway

The paper's central message is that the financial media is a major part of price discovery in some countries and a much weaker public-information channel in others. The cross-country variation is best explained by private informed trading and by the quality of the news transmission mechanism. In markets where insiders appear to trade before announcements, public news arrives partly stale.